AuraOS Paper VIII: Evidence-Ordered Relational Arenas for Governed Cognitive Systems

Compiled Relationship Intelligence, Verified Engineering, Spatial Projection, and Atomic Publication

Dallas Courchene
Long Plain First Nation, Manitoba, Canada
aura.os.q@gmail.com

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https://github.com/dallascourchene-commits/AuraOS

Abstract

This paper extends the seven-paper AuraOS defensive-publication sequence, whose claims N1–N30 described the system's polysynthetic finite-state and vector-symbolic foundations, holographic and swarm mechanisms, liquid-network addressing, memristive and rendering paths, finite-state routing, topology, and bounded self-healing protocols. We disclose a later architectural stratum that converts those primitives into an evidence-ordered cognitive operating substrate. The disclosed system compiles an objective into a lexical address, a guarded six-slot route, exact capability and source grounding, a revocable Arena, bounded tools or replaceable workers, staged proposals, independent verification, and an externally authorized decision. The architecture separates planning, authority, proof, presentation, learning, and publication rather than collapsing them into a single model response.

The principal new contributions are: (i) a compiled relational anatomy index and an eight-dimensional Architecture Relationship Atlas that distinguish exact wiring, overlap, adjacency, missing motifs, candidate connections, and prohibited connections; (ii) an Emergent Evidence Spine that converts capability discovery into exact, bounded investigation packets without granting mutation authority; (iii) an Arena Evidence Contract and a policy/identity/lease-bound Gate for verified engineering with external agents; (iv) a graph-guided diagnostic breadboard with typed review lessons and independent replay; (v) proposal-only experience learning, content-addressed temporal continuity, and governed model routing; (vi) a representation-independent Spatial Arena that admits bounded mesh, point-cloud, and Gaussian-splat assets while retaining domain truth outside the renderer; and (vii) an atomic GitHub publication contract using exact-head compare-and-swap while withholding merge authority.

We formalize twenty new defensive prior-art declarations, N31–N50. The claims are deliberately combination-scoped: standard components such as OIDC, MCP, A2A, glTF, Gaussian splatting, finite-state transducers, static analysis, and Git are not claimed in isolation. The novelty asserted is the specific evidence ordering, typed contracts, lifecycle separation, cross-plane composition, authority boundaries, and reproducible implementation disclosed here.

Keywords: cognitive operating systems; neuro-symbolic systems; finite-state routing; capability security; human-in-the-loop governance; software engineering agents; relationship graphs; spatial computing; Gaussian splatting; provenance; defensive publication.

1 Status, Scope, and License

This document is an arXiv-ready defensive-publication draft and a technical specification of implemented and explicitly bounded architecture in the AuraOS repository through the merge of the Aura Architecture Relationship Atlas on July 20, 2026. It is not a legal opinion, patentability determination, safety certification, production-readiness certification, or claim of exclusive ownership over the standard technologies cited in Section 19.

The text and figures of this paper are released under the MIT License. Referenced source code remains governed by the license attached to the corresponding repository snapshot. A reader may implement the ideas described here without treating a paper claim as operational authorization inside AuraOS; Aura's own runtime authority is carried only by the exact contracts and governance mechanisms described below.

2 Continuity With Papers I–VII

The previous seven papers establish claims N1–N30. Table 1 defines the exclusion boundary used in this paper. Paper VIII does not restate those claims as new inventions; it treats them as inherited substrate and discloses the later architectural combinations built above them.

Table 1: Seven-paper baseline and exclusion boundary.

Paper	Published focus	Claims already covered	Record
I	Foundation	N1–N8: polysynthetic LLM egress; dual linguistic cortex; sparse sweeps; QDKT; visual topology; zero-downtime hot-swap; edge constraints.	Zenodo 20635424
II	Holographic swarm systems	N9–N13: holographic headers; gas-free fractal ledger; swarm learning; VSA rendering; finite-state narrative mechanisms.	Zenodo 20657391
III	Liquid Internet	N14: VSA-addressed routing and naming without dependence on conventional cognitive-layer IP/DNS tables.	Zenodo 20659314
IV	Memristive and rendering upgrades	N15–N17: memristive hyper-epochs; timestep-aware quantization; Gaussian/VSA rendering dynamics.	Zenodo 20673206
V	FST routing and self-refactoring	N18–N20: finite-state routing core; topology resonance; bounded self-refactoring incubator.	Zenodo 20681601
VI	Enhanced FST and topology	N21–N23: FST lexicon; resonance topology; finite-state impact analysis.	Zenodo 20682051
VII	Protocol-layer innovations	N24–N30: protocol invariants; micro-module crystallization; resonant tests; thermal-cost arbitration; deterministic compression; local VSA compute mesh; bounded self-healing.	Zenodo 20695562

2.1 What Paper VIII Adds

The post-Paper-VII architecture is not merely a larger list of modules. It introduces a constitutional organization of those modules into distinct planes with explicit ownership and authority. Its central contribution is a rule that can be expressed compactly:

$$\boxed{\text{Planning proposes. Governance authorizes. Verification proves.}} \tag{1}$$

Presentation, semantic similarity, topology, model output, research, compression, and learned heuristics may aid discovery or explanation. None of them can manufacture authority that is absent from exact evidence and an authorized decision path.

3 Design Premises

3.1 Evidence-Ordered Truth

Let the evidence class of an assertion x be $E(x) \in \{E_1, E_2, E_3, E_4\}$, where:

$$E_1 : \text{exact executable gates, tests, replay, and verifier disposition,} \tag{2}$$
$$E_2 : \text{deterministic comparative proxies with comparable quality,} \tag{3}$$
$$E_3 : \text{explicitly estimated structural projections,} \tag{4}$$
$$E_4 : \text{discovery scans and candidate hypotheses.} \tag{5}$$

The permitted claim relation is monotone only downward:

$$E_1 \succ E_2 \succ E_3 \succ E_4, \qquad E_i \not\Rightarrow E_j \text{ for } i > j. \tag{6}$$

Thus, a topology scan may identify a candidate capability, but it cannot establish that the capability is implemented. A deterministic proxy may compare two bounded workflows, but it cannot be presented as provider billing. Unknown usage remains unknown rather than becoming zero.

3.2 Hard Guards Before Soft Ranking

For a candidate operation c, Aura evaluates a conjunction of hard guards before any advisory ranker is allowed to act:

$$G(c) = G_{\text{state}}(c) \wedge G_{\text{cap}}(c) \wedge G_{\text{policy}}(c) \wedge G_{\text{lease}}(c) \wedge G_{\text{consent}}(c) \wedge G_{\text{fresh}}(c) \wedge G_{\text{risk}}(c) \wedge G_{\text{verify}}(c). \tag{7}$$

The decision rule is:

$$\text{admit}(c) = \begin{cases} 0, & G(c) = 0, \\ 1, & G(c) = 1, \end{cases} \qquad \text{rank}(c) \text{ is evaluated only if } \text{admit}(c) = 1. \tag{8}$$

No high VSA resonance, language-model confidence, sensor score, visual salience, or heuristic utility can rescue a failed hard guard.

3.3 Canonical Ownership

Each consequential truth has one canonical owner and may have many bounded projections. If O_d is the canonical owner for domain d and $P_{d,k}$ is a projection, the architecture requires:

$$P_{d,k} = \mathrm{proj}_k(O_d, \pi_k), \qquad \mathrm{authority}(P_{d,k}) \subseteq \mathrm{authority}(O_d), \tag{9}$$

where π_k is a purpose, privacy, and evidence policy. A projection may reduce authority but cannot enlarge it. This principle governs the Observatory, Spatial Arena, domain dashboards, compatibility formats, checkpoints, and external-agent packets.

4 End-to-End Cognitive Operating Flow

Figure 1 presents the canonical operating flow. The system is designed to accept ordinary language while retaining deterministic internal addresses, bounded execution contexts, and explicit proof obligations.

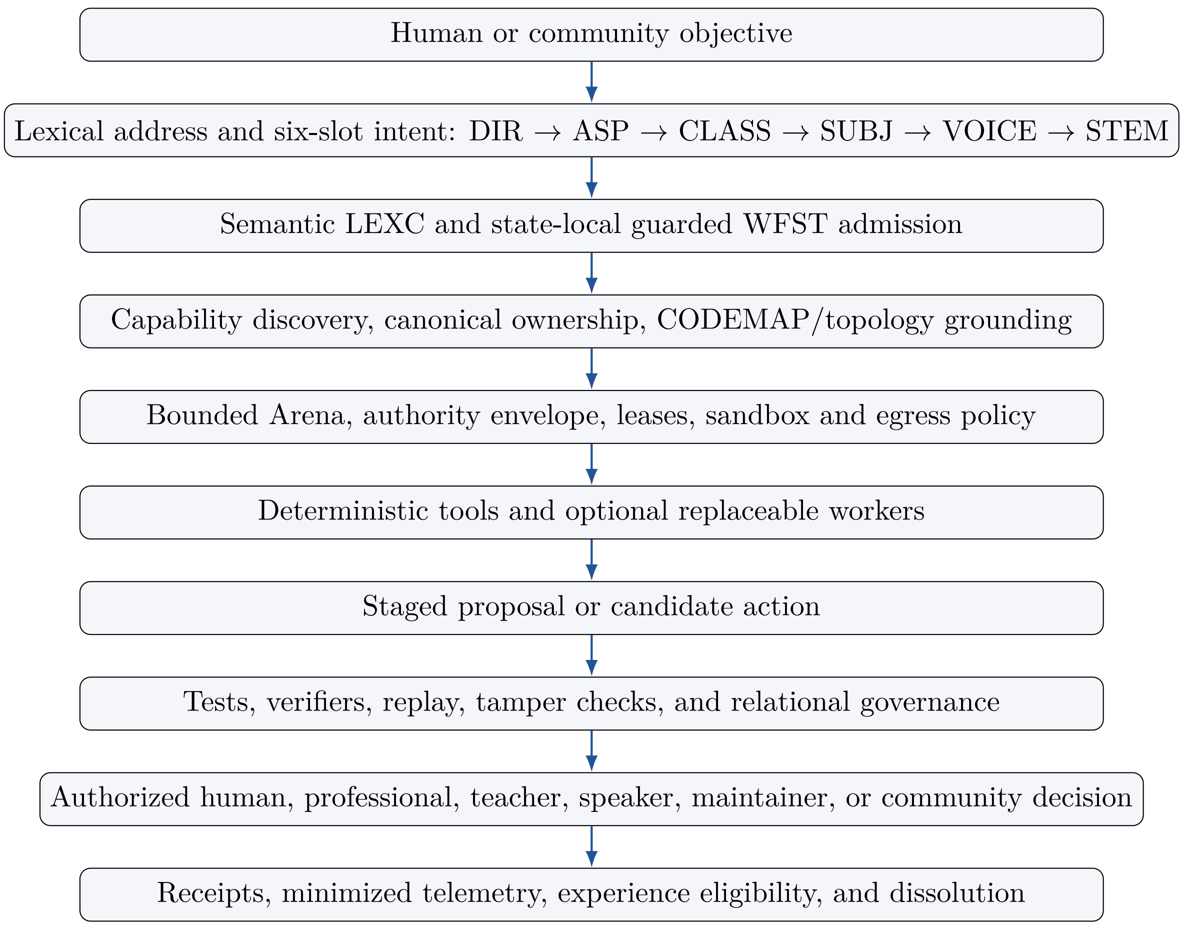

Figure 1: AuraOS evidence-ordered operating flow. Each stage narrows or preserves authority; no stage silently promotes advisory evidence.

5 Compiled Self-Model and Relational Intelligence

5.1 Repository Self-Model

Aura's self-model is compiled from current repository evidence rather than model memory. The synchronized CODEMAP snapshot used for this disclosure reports 1,270 files, approximately 4.24 million estimated text tokens, 9,971 topology nodes, and 22,883 topology edges. Symbol identities, signatures, hashes, line ranges, imports, calls, schemas, tests, routes, command indexes, and neighbor relations are maintained as navigation evidence. Exact source spans and hashes remain patch authority.

The Capability Connectome and Capability Genome Resolver answer a different question: which canonical capabilities already exist and can be reused for an objective? This reuse-first rule reduces duplicate systems and makes ownership drift detectable.

5.2 Ahead-of-Time Relational Anatomy Index

Let V be typed participants and R be typed relations over the exact CodeTopo graph and advisory capability declarations. The Relational Anatomy Index is a deterministic materialization:

$$I = \mathrm{AOTIndex}(V, R, M, F), \tag{10}$$

where M is macro-domain and surgical-bundle metadata and F is a freshness identity containing repository, CODEMAP, topology, Connectome, atomic-inventory, schema, and ontology digests. A real repository build reported 12,357 participants, 24,510 relations, 22 groups, 779 Python source files, and 10,241 atomic callables. Two complete builds were byte-identical. The generated index is a navigation and analysis cache with `safe_to_patch=false`.

5.3 Architecture Relationship Atlas

The Aura Architecture Relationship Atlas (AARA) is a compiled status and intelligence plane over the Relational Anatomy Index. It does not duplicate source truth. For each relationship assessment a, the Atlas retains an eight-dimensional tuple:

$$\Phi(a) = \langle S, M, W, R, L, T, A, P \rangle, \tag{11}$$

where:

- S: structural status (exactly wired, declared, partial, temporary, stale, unwired, unresolved);
- M: semantic relationship (direct, overlapping, complementary, auxiliary, adjacent, redundant, competing, unrelated, contradicted);
- W: wiring disposition (required, recommended, optional, candidate, deferred, not needed, prohibited);
- R: readiness (ready or missing grounding, role, schema, test, verifier, authority, consent, lease, research, or safety);
- L: lifecycle (persistent, objective-scoped, leased, ephemeral, dormant, deprecated, dissolved);
- T: truth class (exact source/schema/test/manifest/runtime or advisory/inferred/unresolved);

- A: authority posture (read-only, advisory, proposal-only, verifier/human/community/lease required, allowed, prohibited);
- P: proof status (open, satisfied, contradicted, deferred).

This non-flattened model prevents a semantically attractive relationship from being misrepresented as exact wiring or execution permission.

5.4 Motifs, Prohibitions, and Delta Receipts

A motif m is a typed role template. If B_m is the set of required roles and $b_m \subseteq B_m$ is the set observed in current evidence, the missing-role set is:

$$\Delta_m = B_m \setminus b_m. \tag{12}$$

The implemented registry covers input-to-authority, state lifecycle, review-packet integrity, external-agent lease, learning-to-reproof, spatial explanation, and cross-Arena adapter motifs. Separately, prohibition patterns detect affinity-based mutation, self-verification, agent self-upgrade, circular authorization, leaked ephemeral leases, direct research-to-production coupling, and unauthorized cross-Arena coupling.

Candidate ranking is constrained by prohibition:

$$C = \{a \mid W(a) = \text{CANDIDATE} \wedge a \notin \mathcal{B}\}, \tag{13}$$

where $\mathcal{B}$ is the active prohibition set. Rebuilding the Atlas yields a content-addressed snapshot and, when a previous snapshot exists, a delta receipt binding the previous and current digests.

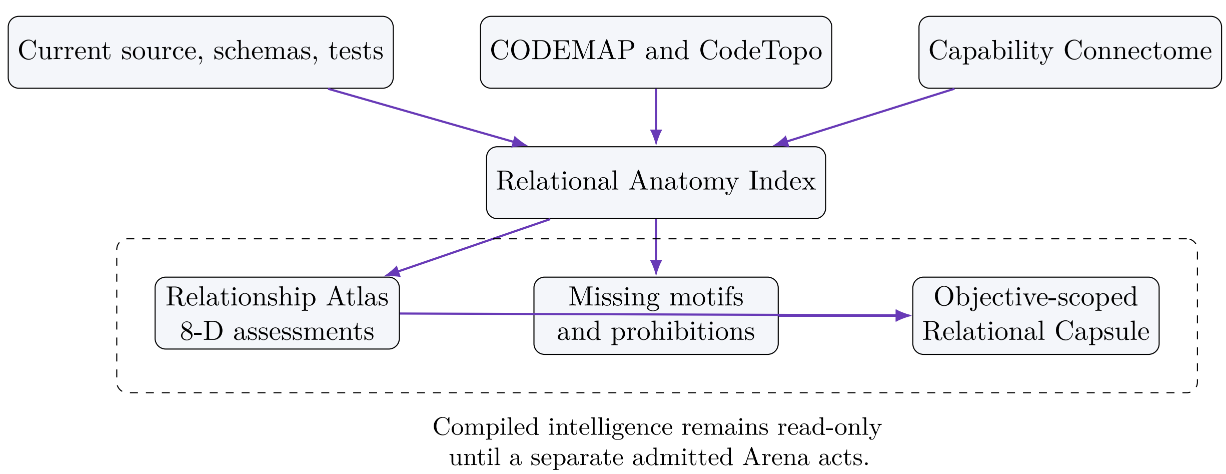

Figure 2: Relational self-understanding without duplicate truth ownership.

6 Emergent Evidence Spine

The Emergent Properties subsystem is not permitted to convert novelty hypotheses directly into code or policy. Instead, the Emergent Evidence Spine turns broad discovery into a bounded evidence workflow:

$$\begin{aligned} &\text{Connectome} \to \text{Atomic Inventory} \to \text{CodeTopo Closure} \\ &\qquad \to \text{Emergent Auditor} \to \text{Research} \to \text{Arena Packet}. \end{aligned} \tag{14}$$

The atomic inventory identifies callable units and their current digests. CodeTopo closure identifies exact callers, callees, imports, tests, schemas, and shared-resource neighbors. The auditor labels missing connections, possible combinations, and contradictions. Research can strengthen or weaken a hypothesis but remains a sidecar. Only a separately admitted Arena can stage a change, and that change must be proven against exact current source.

This architecture supports a useful distinction:

$$\text{emergent candidate} \neq \text{implemented capability}, \tag{15}$$

$$\text{implemented capability} \neq \text{authorized operation}, \tag{16}$$

$$\text{authorized operation} \neq \text{verified success}. \tag{17}$$

7 Arena Runtime, Planning, and Relational Authority

7.1 Arena Lifecycle

An Arena is a bounded context with state identity, route identity, capability leases, declared tools, sandbox and egress policy, verifier requirements, receipts, and explicit authority limits. An ephemeral organ is a temporary capability system compiled for one objective:

$$O_q = \text{compile}(q, I_q, D_q, S_q, L_q, V_q), \tag{18}$$

where q is the objective, I_q its exact inputs, D_q the dependency closure, S_q the sandbox/runtime, L_q the leases and boundaries, and V_q the required verifiers. The organ must dissolve after completion, denial, expiry, or failure, producing a receipt that records what was and was not released.

7.2 Planning Board

The Planning Board is a proposal-only intermediate representation over goals, actions, predicates, constraints, effects, contingencies, variables, and continuity stages. It supports bounded backward regression and forward replay:

$$G_{t-1} = \text{regress}(G_t, a_t), \tag{19}$$

$$S_{t+1} = \text{apply}(S_t, a_t), \tag{20}$$

subject to explicit preconditions, effects, no-progress detection, cycle detection, and frontier convergence. A valid plan cannot authorize itself.

7.3 Relational Authority

An authorization envelope α binds permission to an exact action or event:

$$\alpha = \langle id, \mathrm{H}(a), cap, policy, role, delegation, quorum, t_0, t_1, risk, V, disposition \rangle. \tag{21}$$

Emergency authority remains narrower than ordinary authority, is time-bounded, reason-bearing, and review-producing. Governance may admit an action into an Arena; only verifiers establish whether it succeeded.

8 Verified Engineering Operating System

8.1 Aura Forge and the Arena Evidence Contract

Aura Forge is a product-level facade over existing Coding Arena owners. It freezes an engineering objective into an `AURA_FORGE_ARENA_EVIDENCE_CONTRACT_V1` containing repository identity, CODEMAP identity, plan phase, source references, allowed files, dependencies, tests, risk map, budgets, authority limits, and required gates. The workflow is:

$$\text{FRAME} \rightarrow \text{GROUND} \rightarrow \text{PLAN} \rightarrow \text{ACT} \rightarrow \text{PROVE} \rightarrow \text{DECIDE} \rightarrow \text{DISSOLVE}. \tag{22}$$

A replaceable worker receives only a source/test slice lease and returns a bounded diff or edit plan. Forge stages and verifies the candidate and stops at `READY_FOR_HUMAN_REVIEW`. It cannot commit, push, create a pull request, merge, or mutate production.

8.2 Selective Council and Sliced Surgeon

Architecture-level deliberation is separated from exact-file implementation. Selective Council V3 activates only critic lanes justified by plan structure, dependency, continuity, risk, or uncertainty. The Sliced Surgeon receives exact files, symbols, tests, and local constraints. Local assertion failures may return to the Surgeon; invalidated interfaces, dependencies, invariants, authority assumptions, or materially expanded scope return to the Council.

Grounded phase capsules allow multiple phases to share one exact evidence bundle without exposing unrestricted repository context to every worker. The refactor output record preserves prompt/patch identity, exact gates, failures, disposition, usage class, and claim boundaries.

8.3 Aura Gate

Aura Gate wraps an exact retained Forge contract with authenticated identity, static policy, an expiring Arena lease, governed egress, append-only audit evidence, and bounded protocol adapters. The trust chain is:

$$\begin{aligned} \text{OIDC verify} &\rightarrow \text{policy admit} \rightarrow \text{Forge prepare} \\ &\rightarrow \text{authority envelope} \rightarrow \text{lease} \\ &\rightarrow \text{pre-action audit} \rightarrow \text{exact start.} \end{aligned} \tag{23}$$

Identity comes from a verified transport boundary, never from model output, MCP arguments, or A2A message bodies. Every outbound payload is bound to purpose, destination, data class, retention, exact bytes, and budget. Cancellation is explicit lease revocation. The current implementation is a private, loopback-oriented Forge-specific proof and does not claim to be a complete production OAuth/MCP/A2A deployment.

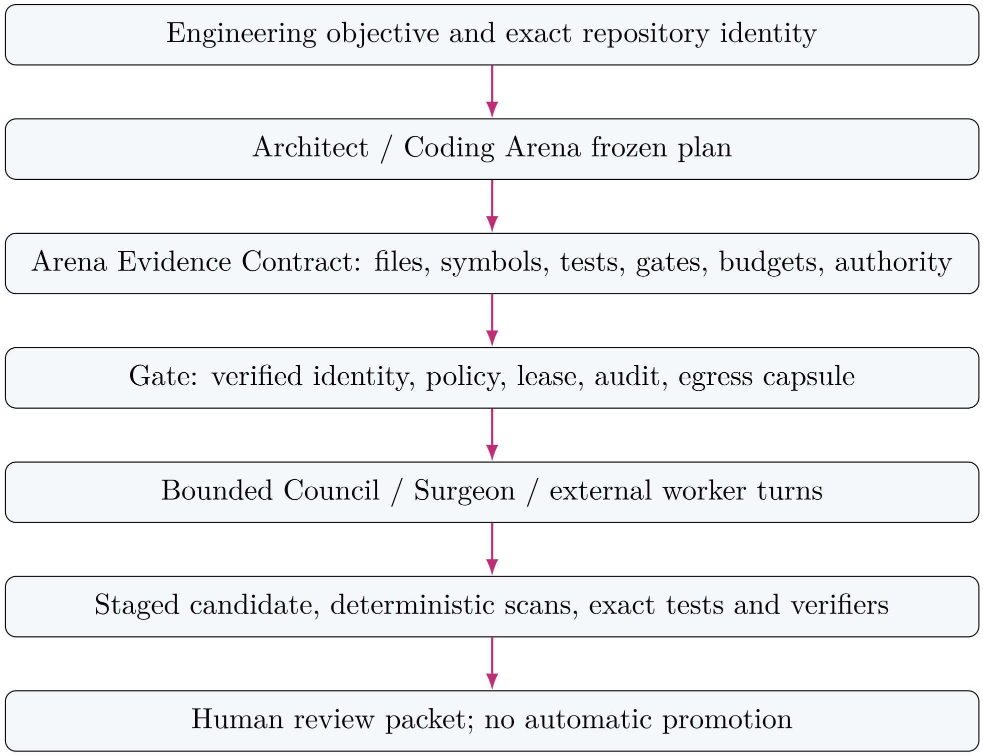

Figure 3: Evidence-bound engineering and identity/policy-bound external-agent execution.

9 Graph-Guided Review and Review Learning

9.1 Coding Waboose

Coding Waboose is a separate pre-repair review organ with lifecycle:

$$\begin{aligned} \mathrm{FRAME} \to \mathrm{DIFF} \to \mathrm{SLICE} \to \mathrm{SCAN} \\ &\to \mathrm{INVESTIGATE} \to \mathrm{CORROBORATE} \to \mathrm{RANK} \\ &\to \mathrm{DECIDE} \to \mathrm{DISSOLVE}. \end{aligned} \tag{24}$$

Aura computes changed files and symbols, callers, callees, tests, schemas, shared-resource neighbors, exact source anchors, deterministic findings, evidence status, deduplication, and precision-first ranking. A coding model may propose focus directives or semantic findings but cannot invent authoritative edges or mark its own finding proven.

9.2 Diagnostic Breadboard

The Planning Board is specialized into a diagnostic circuit board. Each suspected defect is connected to explicit proof paths: forward effects through callees and schemas, backward prerequisites through callers and authority contracts, and lateral shared-resource or state-transition edges. A finding is reportable as confirmed only when exact current source or an executable verifier corroborates it.

9.3 Typed Lessons and Crucible Replay

External reviews from Codex, CodeRabbit, or humans can be normalized into typed lessons. Deterministic detectors then search for defect classes such as authority aliases, protected metadata overrides, count-only bounds, unsafe source paths, URI aliases, schema/runtime drift, unwired regressions, stale evidence, and source-integrity failures. Lessons are teacher signals, not self-modifying rules. They are replayed against exact repository ancestry through Crucible, and a clean replay is evidence only for the reviewed head.

10 Model Cognome and Empirical Cost Governance

The Model Cognome separates endpoint identity, observed capability, benchmark class, telemetry, cost completeness, privacy/egress policy, route proposal, live authorization, drift, quarantine, federation, promotion, and rollback. Supported route classes include zero-model, direct, cascade, and panel. Operating modes include legacy, shadow, and paired-live.

A paired-live authorization binds purpose, graph digest, endpoint, verifier, expiry, budget, nonce/replay policy, egress class, rollback, and quarantine. A shadow route cannot call a provider. A higher shadow score cannot promote itself.

The Empirical Cost Observatory preserves distinct quantities:

$$U = \langle u_{provider}, u_{tokenizer}, u_{proxy}, b, c_{measured}, c_{calculated}, c_{estimated}, c_{unavailable} \rangle. \tag{25}$$

The system may compute cost per verified success, but an estimate cannot be re-labeled as an invoice. This separation is part of the control architecture, not merely reporting style.

11 Verified Experience, Proposal-Only Learning, and Temporal Continuity

11.1 Experience Pipeline

A raw prompt, model response, research paper, failed attempt, or route trace is not learned truth. The eligible pipeline is:

$$\begin{aligned} \text{governed execution} &\to \text{verifier evidence} \to \text{OutcomeVector} \\ &\to \text{ArenaExperience V3} \\ &\to \{\text{TRAIN}, \text{VALIDATION}, \text{SHADOW}\} \\ &\to \text{CRYSTALLIZATION\_PROPOSED}. \end{aligned} \tag{26}$$

The current permitted crystallization surface is intentionally narrow. Crucible cannot alter hard guards, transitions, capabilities, consent, risk classes, source code, active grammar, production model routes, civic decisions, financial truth, construction truth, or repository publication state.

11.2 Attempt Archive and Observatory Separation

The Attempt Archive preserves successful, denied, and failed attempts with exact objective, evidence, requested route, denial or admission, worker/tool activity, verifier result, and disposition. Failures are retained for debugging and audit rather than silently deleted. The Observatory explains objectives, routes, evidence, topology, compression, workers, and proof requirements but cannot create a lease, execute a worker, stage a patch, approve a domain action, or make an item eligible for learning.

11.3 Temporal Persistence

Temporal continuity consists of compact intra-session state, content-addressed inter-session checkpoints, an append-only parent/fork registry, Arena-specific projections, guard evidence, and a review-only restoration packet. A checkpoint binds repository head, payload digest, invariant digest, parent or fork lineage, and freshness classification.

Restoration yields one of a bounded set of review decisions, such as direct-resume review, mitosis required, or restoration-council required. No checkpoint automatically restores state, calls a model, applies code, promotes grammar, authorizes physical action, or publishes a repository change. Cross-Arena batons carry references and digests rather than unrestricted payloads.

12 Representation-Independent Spatial Arena

12.1 Spatial Truth and Lifecycle

Aura Spatial is a governed presentation plane over canonical domain owners. It introduces immutable frames, assets, entities, links, scenes, interactions, device profiles, render plans, receipts, and dissolution contracts. Its finite lifecycle is:

$$\begin{aligned}\text{FRAME} &\to \text{GROUND} \to \text{COMPILE\_SCENE}\\ &\to \text{PLAN\_RENDER} \to \text{PRESENT} \to \text{INTERACT}\\ &\to \text{PROVE} \to \text{DECIDE} \to \text{DISSOLVE}.\end{aligned} \tag{27}$$

A scene snapshot is a content-addressed projection, not a second domain ledger. Renderer state, browser caches, telemetry, interactions, and checkpoints are not domain authority. `DECIDE` produces a packet for an external authorized decision; it does not apply the decision.

12.2 Bounded Interchange and Gaussian Representations

The interchange layer admits local glTF/GLB and PLY assets under explicit unit, coordinate, provenance, metadata, and resource bounds. External resource fetches and executable content are rejected. The Gaussian path admits bounded local SPZ v4 and the release-candidate `KHR_gaussian_splatting` representation, with deterministic point-cloud fallback, immutable decoded payloads, digest binding, two-pass no-copy preflight, aggregate allocation, GPU and frame budgets, cancellation, and disposal.

Let B_a, B_g, and B_f be aggregate allocation, GPU, and frame budgets. Admission requires:

$$\sum_i bytes_i \le B_a, \qquad \sum_i gpu_i \le B_g, \qquad work_{frame} \le B_f, \tag{28}$$

before the renderer allocates the complete representation. Unsupported Gaussian features degrade to a declared fallback rather than silently changing truth class.

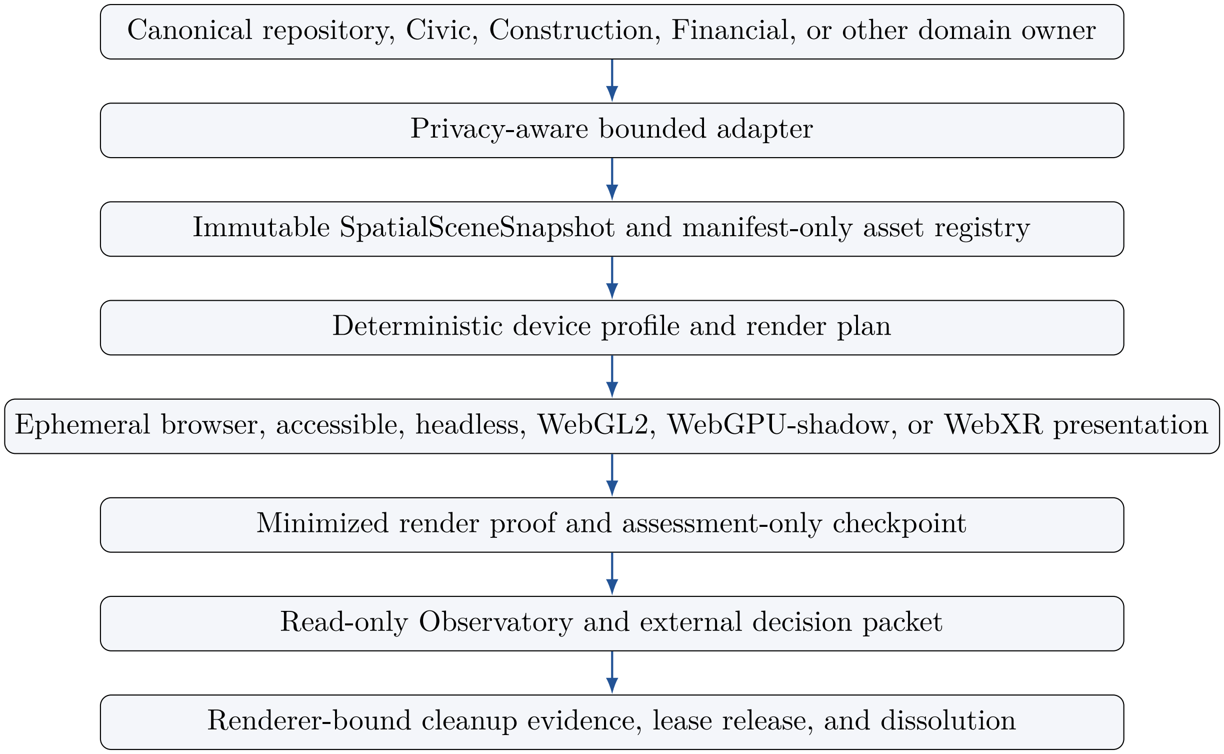

Figure 4: Projection-only Spatial Arena. Visual fidelity and interaction do not transfer domain authority to the renderer.

13 Domain Projection Pattern

13.1 Civic Commons

Civic Commons combines explicit truth classes, official-source snapshots, local community overlays, consent and dissent preservation, needs and assets, decomposition, trade-off surfaces, simulation-only what-if analysis, reversible pilot design, and privacy-filtered memory. It rejects person-level vulnerability mapping, hidden winners, binding votes, automatic funding, legal approval, procurement, surveillance, or service allocation.

13.2 Construction

Construction has one canonical truth owner, `ConstructionProjectState`, reconstructed from immutable claims, evidence, and events. Exact readiness, expiry, conflict, and scope blockers run before any advisory model or sensor ranking. A privacy-minimized Human Agent and Spatial projection can show blocked and admissible candidates, deterministic roles, and proposal-only time/cost/idle deltas. No projection can authorize physical work, payment, access, equipment, inspection, safety, engineering, law, regulation, survey truth, source mutation, or merge.

13.3 Financial Exact State

The Financial exact-state layer uses immutable Decimal-backed records, explicit currency, units, dates, provenance, freshness, and truth classes: user-recorded, imported exact, derived arithmetic, assumption, and unavailable. It rejects silent float coercion, implicit rounding, inferred ownership or consent, implicit currency conversion, contradictory dates, duplicate identities, and model-estimated values presented as exact. It is not an adviser and cannot transact.

Together these domains demonstrate a reusable pattern: preserve canonical domain truth, compile a purpose-limited proposal or projection, keep hard blockers before advisory ranking, and return consequential decisions to the authorized external body.

14 Atomic GitHub Publication Without Merge Authority

The Agent Bridge publication lane replaces temporary workflow payloads with a canonical publication contract and GitHub's server-side commit mutation. It supports create and update modes.

In create mode, the base snapshot is immutable and the branch must be fresh. In update mode, the contract binds the exact open, unmerged, same-repository pull request number and exact current head SHA. A publication is accepted only if the expected head still matches:

$$\text{publish}(C, h_e) = \begin{cases} \text{server-side commit}, & h_{observed} = h_e, \\ \text{fail closed with recovery evidence}, & h_{observed} \neq h_e. \end{cases} \tag{29}$$

The lane validates canonical operation type, path, UTF-8 or bounded Base64 encoding, additions, deletions, file and aggregate byte budgets, same-repository PR identity, and restricted paths. Caller-supplied `.github/workflows/` changes are rejected. Tokens are taken only from the operator environment, never from MCP arguments. Merge preparation returns evidence and explicitly false merge authority; a separately authenticated trusted human performs any merge.

15 Continuity and Compression Substrates

Aura uses several compact representations while preserving their distinct truth classes:

- Context Crusher and exact slicing remove unrelated repository context;
- ST3GG provides compact frames and exact-recall handles, with admission that accounts for protocol overhead;
- JSpace J0/J1/J2 carries compact route, Arena, and cross-system continuity;
- QDKT stores canonical observation events and read-only compatibility evidence;
- Symbolic Trace Memory separates raw references, compact atoms, and independently consolidated canvases;
- State Ledger V3 stores compact intra-session execution state;
- Temporal Persistence stores content-addressed checkpoints and payload-free handoffs.

Compatibility readers may preserve legacy artifacts, but new writes go to canonical owners. Translation must preserve IDs and provenance where possible, label compatibility output, fail closed on unsupported fields, and never transfer ownership back to a legacy format.

16 Security, Privacy, and Cultural Governance

Security is architectural: least-privilege capability leases, exact purpose and expiry, repository confinement, real sandbox requirements for arbitrary code, content-addressed evidence, append-only events, replay/nonce rules, governed egress, local/quarantine/federation boundaries, aggregate-only public telemetry, no secrets in prompts or committed artifacts, and fail-closed unknowns.

Cultural and community governance is not an optional presentation layer. The architecture requires speaker, teacher, professional, or community authority where appropriate; prohibits invented translations and semantic activation of restricted cultural profiles; keeps distinct Nations, languages, and conventional technical influences distinct; and supports consent, purpose limitation, revocation, and deletion paths where the governing contract requires them.

17 Implementation Evidence and Reproducibility

The source snapshot for this disclosure is merge commit `776d1728...477fab`. Table 2 records representative evidence without upgrading it beyond its class.

Table 2: Representative repository evidence.

Artifact or subsystem	**Evidence**	**Claim boundary**
CODEMAP / compiled topology	1,270 files; 9,971 nodes; 22,883 edges	Navigation and current-tree self-model; not patch authority.
Relational Anatomy Index	12,357 participants; 24,510 relations; 22 groups; 10,241 atomic callables; byte-identical full builds; 39 focused tests	Deterministic AOT relationship cache; exact relations remain source-pinned; advisory declarations remain advisory.
Relationship Atlas R0–R1	38 focused tests; eight dimensions; seven prohibitions; seven motif templates; deterministic snapshot and delta receipts	Compiled status plane; no truth, patch, or execution authority.
Spatial S0–S6	Immutable contracts, bounded importers, render plans, sessions, Construction binding, Agent Bridge/MCP/CLI, adversarial asset checks	Projection and review only; no domain mutation or renderer authority.

Artifact or subsystem	Evidence	Claim boundary
Coding Waboose review learning	Typed lessons, deterministic detector classes, source-integrity and ancestry replay	Review leads require exact corroboration; no self-confirmation or automatic repair.
Aura Gate Phase 2 proxy	37,907 input, 1,852 output, 39,759 total proxy; 51,987 estimated saved (56.66%) against documented counterfactual	Derived char/4 engineering proxy, not provider billing and not full-session usage.
Context localization benchmark	89.04% lower total deterministic proxy with quality +0.0057	Controlled comparative fixture, not universal model-performance evidence.
Selective Council V3	32.83% lower proxy than Council V2 with same accepted patch and quality on controlled fixture	Bounded comparative result only.
State Ledger continuity	96.19% less step-7 context; preservation 1.0000; drift 0.0000	Synthetic continuity fixture.

Reproduction begins with the exact repository snapshot, regeneration of CODEMAP/topology, validation of the Relational Index and Atlas, and focused tests for each claim family. Generated maps should be rebuilt rather than trusted from historical line numbers.

18 Defensive Prior-Art Declarations N31–N50

The following declarations are implementation- and combination-scoped. They do not claim the underlying standard primitives in isolation. Each claim refers to the complete combination, including its negative authority boundaries.

18.1 Claim N31: Evidence-Ordered Cognitive Operating Constitution

We declare prior art in a cognitive operating substrate that constitutionally separates proposal, authority, proof, presentation, and learning; orders exact evidence above advisory cognition; evaluates hard guards before ranking; and fails closed on stale, malformed, ambiguous, conflicting, expired, or unauthorized work. The distinctive combination is the use of this constitution as a shared invariant across routing, coding, civic, construction, financial, spatial, learning, and publication planes.

Representative implementation: `.aura/ARCHITECTURE.md`, `aura_event_contracts.py`, `aura_arena_wfst_runtime.py`, authority, verifier, and Arena modules.

18.2 Claim N32: Guarded Six-Slot Arena Fabric With Relational Authority

We declare prior art in compiling ordinary objectives through a lexical address and ordered DIR–ASP–CLASS–SUBJ–VOICE–STEM packet into state-local guarded WFSTs, then binding an admitted action to exact action digest, capability, policy, role, delegation, quorum, validity, risk, verifier requirements, and external disposition. The novelty asserted is the finite-state plus relational-authority composition and its rule that soft semantic ranking cannot override hard admission.

Representative implementation: `aura.lexc`, `aura_arena_wfst_runtime.py`, `aura_relational_authority.py`, `.aura/arena_routes/`.

18.3 Claim N33: Proposal-Only Planning Board With Bidirectional Symbolic Proof Paths

We declare prior art in a typed planning intermediate representation that supports bounded backward regression from goals, forward replay from state, cycle and no-progress detection, continuity stages, immutable event projection, and independent history reconstruction, while remaining incapable of executing or authorizing its plan. The same IR is specialized into coding breadboards and civic planning shadows without transferring canonical ownership.

Representative implementation: `aura_planning_board.py`, `aura_planning_regression.py`, `aura_planning_frontier.py`, `aura_planning_events.py`, coding and civic projectors.

18.4 Claim N34: Compiled Repository Self-Model and Reuse-First Capability Resolution

We declare prior art in combining a current-tree CODEMAP, semantic symbol identities, compiled deep topology, exact source/test localization, Topological Context Anchors, a Capability Connectome, and a Capability Genome Resolver so that the system searches for canonical reusable owners before proposing a new component. Generated vectors and topology are navigation evidence only; exact source spans and hashes remain patch authority.

Representative implementation: `aura_codebase_navigator.py`, `aura_topological_context_anchor.py`, `aura_capability_connectome.py`, `aura_capability_resolver.py`, `aura_capability_resolver_v2.py`.

18.5 Claim N35: Deterministic Relational Anatomy Index and Eight-Dimensional Relationship Atlas

We declare prior art in materializing a freshness-pinned AOT relational anatomy index over exact structural edges and advisory capability declarations, then compiling an eight-dimensional Relationship Atlas that separately classifies structural status, semantic relationship, wiring disposition, readiness, lifecycle, truth class, authority posture, and proof status. The Atlas identifies overlapping-unwired, auxiliary-adjacent, missing-motif, candidate, and prohibited relationships and emits deterministic snapshot and delta receipts without becoming a truth or mutation owner.

Representative implementation: `aura_relational_index.py`, `aura_relationship_atlas.py`, Atlas schemas, receipts, and focused tests.

18.6 Claim N36: Emergent Evidence Spine From Capability Hypothesis to Bounded Arena Packet

We declare prior art in the staged combination of Capability Connectome, complete atomic inventory, exact CodeTopo closure, bounded Emergent Capability Auditor, research sidecars, and objective-specific Arena preparation. The system may discover candidate combinations and missing wires, but the discovery output cannot patch, verify, authorize, learn, publish, or self-promote.

Representative implementation: `aura_emergent_evidence_spine.py`, `aura_emergent_capability_auditor.py`, Connectome, Affordance Directory, research adapters, and Agent Bridge preparation.

18.7 Claim N37: Ephemeral Organ Compiler With Leases, Sandboxes, Receipts, and Mandatory Dissolution

We declare prior art in compiling a temporary capability system from objective, dependency closure, manifest, sandbox/runtime, capability leases, egress policy, verifier set, telemetry policy, and dissolution contract. Arbitrary code fails closed when the required process isolation is unavailable; completion, denial, expiry, or failure produces a receipt and releases leased capability.

Representative implementation: `aura_ephemeral_runtime.py`, `aura_ephemeral_sandbox.py`, `aura_ephemeral_registry_store.py`, Arena tool runtime and manifests.

18.8 Claim N38: Frozen Arena Evidence Contract With Selective Council and Sliced Surgeon

We declare prior art in a verified engineering runtime that freezes objective, repository identity, plan phase, exact source/test references, allowed files, risk, budgets, required gates, and authority into one deterministic contract; uses selective architecture critics only when justified; leases bounded slices to replaceable workers; stages candidate patches; independently verifies them; and stops at human review without promotion authority.

Representative implementation: `aura_forge.py`, Architect/Coding Arena plans, Controlled Refactor Session, external-LLM slice sessions, staging, verifier, output vault, and refactor output records.

18.9 Claim N39: Identity-, Policy-, and Egress-Bound Agent Gateway Over an Exact Frozen Contract

We declare prior art in wrapping a retained engineering contract with offline verified OIDC identity, pseudonymous actor reference, content-addressed static policy, expiring Arena lease, pre-action append-only audit, purpose-limited exact-byte egress capsules, bounded MCP/A2A message projections, explicit revocation, and dissolution. Protocol-body identity and model-provided authorization are never authority.

Representative implementation: `aura_gate.py`, `aura_gate_oidc.py`, `aura_gate_egress.py`, `aura_gate_audit.py`, `aura_gate_adapters.py`, `aura_gate_server.py`.

18.10 Claim N40: Graph-Guided Diagnostic Breadboard With Agent Findings That Cannot Self-Confirm

We declare prior art in a review organ that computes exact diff/symbol impact, callers, callees, tests, schemas, state transitions, authority boundaries, and shared-resource neighbors; compiles these into forward and backward diagnostic circuits; accepts bounded focus directives or semantic findings from external coding agents; corroborates findings against exact source and tools; and produces a precision-first human review or separate repair handoff.

Representative implementation: `aura_coding_waboose.py`, `aura_coding_waboose_breadboard.py`, `aura_review_arena.py`, Waboose CLI/MCP and schemas.

18.11 Claim N41: Typed Review-Lesson Learning With Source-Integrity and Ancestry Replay

We declare prior art in normalizing bounded external and human review evidence into typed defect lessons; scanning deterministic defect classes; binding evidence to canonical UTF-8 content, source digests, exact Git head/tree, signed or integrity-checked registries, and merge ancestry; replaying lessons through a proposal-only learning system; and prohibiting lessons from writing source, confirming themselves, or claiming future-head coverage.

Representative implementation: `aura_coding_waboose_review_lessons.py`, `aura_coding_waboose_review_learning.py`,
`aura_review_lessons_security.py`, Crucible bridge/MCP, registries and harnesses.

18.12 Claim N42: Grounded Phase Capsules and Persistent External-Worker Quality Evidence

We declare prior art in sharing one exact evidence bundle across bounded phases while leasing each external worker only the source, tests, state, constraints, and output schema required for its turn; persisting attempt, prompt, patch, verifier, failure, repair, usage-class, and disposition evidence; and using that evidence to compare worker routes without granting ambient repository or release authority.

Representative implementation: external-LLM slice-session modules, grounded phase capsule modules, Human Agent Attempt Archive, output-quality records, and executable refactor benchmarks.

18.13 Claim N43: Governed Model Cognome With Shadow/Paired-Live Separation and Cost-Class Integrity

We declare prior art in a model-routing substrate that separately records endpoint identity, capability evidence, benchmark class, cost/latency telemetry, usage completeness, privacy and egress eligibility, route proposal, live authorization, replay, drift, quarantine, federation, promotion, and rollback; supports zero-model/direct/cascade/panel classes; separates shadow from paired-live execution; and preserves measured, calculated, estimated, and unavailable cost as non-interchangeable evidence classes.

Representative implementation: `aura_model_cognome_store.py`, `aura_adaptive_model_router.py`, shadow/paired-live modules, probe, telemetry, drift, federation, and Empirical Cost Observatory owners.

18.14 Claim N44: Verified Arena Experience to Proposal-Only Crystallization

We declare prior art in promoting only verifier-backed governed executions into typed Arena-Experience records; separating TRAIN, VALIDATION, and SHADOW datasets; generating an OutcomeVector; producing only `CRYSTALLIZATION_PROPOSED`; and requiring independent verifier and human review before any narrow soft-weight change. Raw prompts, papers, route traces, and model outputs are ineligible by themselves.

Representative implementation: Experience Ledger V3, ArenaExperience contracts, OutcomeVector, Learning Arena, Crucible, validation and proposal modules.

18.15 Claim N45: Content-Addressed Temporal Continuity With Review-Only Restoration and Payload-Free Cross-Arena Handoffs

We declare prior art in combining compact intra-session state, content-addressed checkpoints, exact repository-head binding, invariant and payload digests, append-only parent/fork lineage, freshness and branch-offset classification, Arena-specific projections, review-only restoration decisions, and cross-Arena batons that carry references and digests rather than payloads. Checkpoints cannot automatically restore or authorize downstream action.

Representative implementation: `aura_temporal_persistence.py`, persistence CLI/bridge, checkpoint registries, State Ledger V3, JSpace continuity, Arena adapters.

18.16 Claim N46: Representation-Independent Spatial Arena With Governed Render and Dissolution Lifecycle

We declare prior art in projecting canonical domain state into immutable, content-addressed frames, assets, entities, links, scenes, interactions, device profiles, and render plans; negotiating bounded accessible/headless/WebGL2/WebGPU-shadow/WebXR presentation; preserving exact scene/session/render bindings; emitting minimized proof and decision packets; and requiring renderer-bound cleanup, lease release, and dissolution without granting renderer or domain-mutation authority.

Representative implementation: `aura_spatial_contracts.py`, coordinate frames, asset registry, scene compiler, render planner, session, Arena, browser adapters, Agent Bridge, MCP, CLI and route grammar.

18.17 Claim N47: Bounded Gaussian and Interchange Admission With Deterministic Fallback and Aggregate Resource Proof

We declare prior art in a local-only spatial importer that preflights glTF/GLB, PLY, SPZ, and `KHR_gaussian_splatting` assets; binds decoded representation digests; enforces explicit units, coordinate conversion, provenance, metadata and external-resource restrictions; performs aggregate allocation/GPU/frame budgeting before complete allocation; provides deterministic point-cloud fallback; and binds cancellation and disposal evidence to the presentation lifecycle.

Representative implementation: spatial interchange and Gaussian importer modules, Python/JavaScript parity paths, adversarial asset corpus, renderer and cleanup receipts.

18.18 Claim N48: Canonical-Domain-to-Bounded-Projection Pattern Across Civic, Construction, and Financial Arenas

We declare prior art in a reusable cross-domain architecture where one exact domain owner feeds a privacy- and purpose-limited proposal or presentation adapter; hard truth-class and authority blockers execute before soft ranking; projections preserve dissent, uncertainty, unavailable values, and blocked options; and consequential decisions remain with an external authorized human, professional, owner, or community. The pattern is instantiated without duplicating domain ledgers in Civic, Construction, Financial, Human Agent, Observatory, or Spatial surfaces.

Representative implementation: Civic Commons, Construction contracts/state/authority/adapters, Financial exact-state contracts, domain Human Agent/Observatory/Spatial projectors.

18.19 Claim N49: Exact-Head Atomic GitHub Publication Contract With Evidence-Only Merge Preparation

We declare prior art in combining an immutable repository snapshot, bounded canonical additions/deletions, exact branch/PR/head identity, server-side multi-file commit creation with expected-head compare-and-swap, no force update, no automatic branch deletion after ambiguous failure, strict path/encoding/budget restrictions, token isolation from agent arguments, recovery evidence, and a separate merge-preparation operation that explicitly lacks connector and merge authority.

Representative implementation: `aura_agent_arena_github_bridge.py`, GitHub MCP projection, publication contracts, recovery and merge-preparation evidence, focused tests.

18.20 Claim N50: Truth-Class-Preserving Multi-Representation Continuity and Compatibility Fabric

We declare prior art in composing exact slicing, ST3GG frames and recall handles, JSpace continuity packets, QDKT observation events, Symbolic Trace Memory, State Ledger, Tensor Evidence propagation, and temporal checkpoints while explicitly preventing any compact, vector, inferred, compatibility, or visual representation from becoming a second truth owner. New writes target canonical owners; legacy reads remain labeled, provenance-preserving, and removable only after measured migration evidence.

Representative implementation: ST3GG V2, JSpace J0/J1/J2, QDKT observation/compatibility modules, Symbolic Trace Memory, Tensor Evidence Arena adapters, State Ledger and migration policies.

19 Related Work and Non-Claimed Primitives

AuraOS builds on well-established work in finite-state transducers and morphology [1–3], vector-symbolic and hyperdimensional representations [4–6], capability security [7, 8], static program analysis and code property graphs [9, 10], planning [11, 12], software-supply-chain provenance [13, 14], identity and agent protocols [15–17], GitHub's commit-authoring mutation [18], glTF and Gaussian splatting [19–21], and human-centered risk governance [22].

The claimed combinations differ from these primitives in several ways:

Table 3: Standard primitive versus AuraOS combination.

Primitive	Typical role	AuraOS combination disclosed here
WFST / FST	Lexical or state-machine recognition/transduction	Six-slot objective address, state-local admission, relational authority, leases, verifier obligations, and hard-guard-before-ranking across multiple Arenas.

Primitive	**Typical role**	**AuraOS combination disclosed here**
Knowledge graph / code graph	Store or query entities and relationships	Exact/advisory split, freshness-pinned anatomy index, eight-dimensional Atlas, motif completion, prohibition registry, delta receipt, and no mutation authority.
Static analysis	Identify code relations or vulnerabilities	Diagnostic breadboard combining exact diff closure, forward/backward proof paths, external-agent focus, source corroboration, repair separation, and review-learning replay.
Capability security	Limit authority through references/capabilities	Objective-scoped Arena organs, expiring leases, relational authorization, sandbox and egress contracts, mandatory dissolution, and domain-specific decision gates.
OIDC / MCP / A2A	Identity or interoperable agent communication	A verified identity boundary that binds exact frozen engineering contract, static policy, lease, pre-action audit, exact bytes, egress capsule, cancellation/revocation, and human review.
Event sourcing / provenance	Reconstruct history and verify artifacts	Planning-event projection, Attempt Archive, source ancestry, content-addressed checkpoints, experience eligibility, and claim-class-preserving audit.
glTF / Gaussian splatting	Exchange and render spatial assets	Local-only admission, units/coordinates/provenance, no external resources, aggregate budgets, deterministic fallback, review-only interaction, domain truth retention, and renderer-bound dissolution.
Git commit APIs	Create repository commits	Exact base/PR/head contract, expected-head CAS, restricted canonical change set, bounded encodings, ambiguous-failure recovery evidence, and merge-preparation with explicit false authority.

20 Limitations and Open Programs

The architecture is extensive but its evidence is bounded. Current tests prove exact artifacts and fixtures, not universal correctness. Token proxies are not provider invoices. Spatial import and rendering support do not establish production XR safety, sensor truth, photogrammetric accuracy, or surveying authority. The Gate is a private Forge-specific proof, not a complete enterprise identity or transport deployment. The Atlas can reveal candidates and prohibitions but cannot prove that every useful relation has been discovered. Domain Arenas do not yet provide real financial, construction, civic, medical, legal, or cultural operational authority.

Architecture-supported but separately gated future programs include sovereign Arena federation, intent-compiled consumer application fabrics, disaster coordination, institutional deployments, production WebXR/OpenXR, capture and reconstruction, remote rendering, additional governed Spatial adapters, real owner/contractor/payment/access/sensor connectors, financial scenarios and connectors, module marketplaces, and production promotion. These directions are not represented as implemented claims in this paper.

21 Conclusion

AuraOS has evolved from a collection of edge-native, finite-state, vector-symbolic, topology, rendering, and self-healing mechanisms into an evidence-ordered cognitive operating architecture. The key architectural move is not unrestricted autonomy; it is the systematic refusal to let meaning, similarity, planning, models, visualizations, checkpoints, research, or learned heuristics impersonate truth or authority.

Paper VIII discloses twenty combination-scoped prior-art claims, N31–N50, covering the constitutional evidence order, guarded Arena fabric, proposal-only planning, compiled self-model, Relational Anatomy Index, Relationship Atlas, Emergent Evidence Spine, ephemeral organs, verified engineering, Gate, Waboose, review learning, external-worker evidence, Model Cognome, verified experience, temporal continuity, governed spatial projection, bounded Gaussian interchange, domain projection, atomic publication, and multi-representation continuity. Together they describe a reproducible pattern for building cognitive systems that can discover widely, act narrowly, prove explicitly, and remain governed by the humans and communities whose authority they are meant to serve.

A Claim-to-Artifact Crosswalk

Claim	Primary architecture	Representative proof surfaces
N31–N33	Constitution, guarded routing, Planning Board, relational authority	Architecture invariants, route manifests, planning regression/replay tests, authority contracts.
N34–N36	CODEMAP, Connectome, Relational Index, Atlas, Emergent Evidence Spine	Current-tree maps, deterministic index/Atlas builds, receipts, schemas, 39+38 focused tests.
N37–N39	Ephemeral organs, Forge, Council/-Surgeon, Gate	Sandbox/lease tests, Arena Evidence Contract, OIDC/policy/egress/audit tests, human-review terminal state.
N40–N42	Waboose, breadboard, lessons, source integrity, slice sessions	Exact diff closure, detector and replay harnesses, ancestry/source digests, persistent output records.
N43–N45	Model Cognome, cost evidence, Experience/Crucible, persistence	Shadow/paired-live gates, usage-class records, dataset separation, checkpoint/-fork/tamper tests.

Claim	Primary architecture	Representative proof surfaces
N46–N48	Spatial Arena, Gaussian/import paths, Civic/Construction/Financial projection	S0–S6 schemas and adversarial assets, route validation, sole-truth-owner and no-authority tests.
N49–N50	Atomic GitHub publication and continuity/compatibility fabric	Exact-head CAS tests, restricted paths/encodings, false merge authority, labeled compatibility and migration tests.

B Suggested arXiv Metadata

- **Primary category:** cs.AI.
- **Secondary categories:** cs.SE, cs.HC, cs.CY, cs.DC.
- **Comments:** Continuation of AuraOS defensive-publication sequence; 20 new combination-scoped claims N31–N50; source snapshot and reproducibility information included.
- **License for paper:** MIT License.
- **Repository snapshot:** `776d1728698f68b19762f76e73b7d9f024477fab`.

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References

[1] K. R. Beesley and L. Karttunen, *Finite State Morphology.* CSLI Publications, 2003.

[2] M. Mohri, "Finite-state transducers in language and speech processing," *Computational Linguistics*, vol. 23, no. 2, pp. 269–311, 1997; updated tutorial treatments 2002.

[3] C. Allauzen, M. Riley, J. Schalkwyk, W. Skut, and M. Mohri, "OpenFst: A general and efficient weighted finite-state transducer library," in *Proc. CIAA*, 2007, pp. 11–23.

[4] T. A. Plate, "Holographic reduced representations," *IEEE Transactions on Neural Networks*, vol. 6, no. 3, pp. 623–641, 1995.

[5] P. Kanerva, "Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors," *Cognitive Computation*, vol. 1, no. 2, pp. 139–159, 2009.

[6] D. Kleyko et al., "Vector symbolic architectures as a computing framework for emerging hardware," *Proceedings of the IEEE*, vol. 110, no. 10, pp. 1538–1571, 2022.

[7] J. B. Dennis and E. C. Van Horn, "Programming semantics for multiprogrammed computations," *Communications of the ACM*, vol. 9, no. 3, pp. 143–155, 1966.

[8] M. S. Miller, K.-P. Yee, and J. Shapiro, "Capability myths demolished," Technical Report SRL2003-02, Johns Hopkins University, 2003.

[9] F. Yamaguchi, N. Golde, D. Arp, and K. Rieck, "Modeling and discovering vulnerabilities with code property graphs," in *IEEE Symposium on Security and Privacy*, 2014.

[10] P. Avgustinov et al., "QL: Object-oriented queries on relational data," in *Proc. ECOOP*, 2016.

[11] R. E. Fikes and N. J. Nilsson, "STRIPS: A new approach to the application of theorem proving to problem solving," *Artificial Intelligence*, vol. 2, no. 3–4, pp. 189–208, 1971.

[12] M. Ghallab et al., "PDDL – The Planning Domain Definition Language," Yale Center for Computational Vision and Control, Technical Report CVC TR-98-003/DCS TR-1165, 1998.

[13] S. Torres-Arias, H. Afzali, T. K. Kuppusamy, R. Curtmola, and J. Cappos, "in-toto: Providing farm-to-table guarantees for bits and bytes," in *USENIX Security Symposium*, 2019.

[14] J. Samuel, N. Mathewson, J. Cappos, and R. Dingledine, "Survivable key compromise in software update systems," in *ACM CCS*, 2010.

[15] N. Sakimura, J. Bradley, M. Jones, B. de Medeiros, and C. Mortimore, "OpenID Connect Core 1.0," OpenID Foundation, Feb. 2014.

[16] Model Context Protocol, "Specification revision 2025-06-18," 2025. `https://modelcontextprotocol.io/specification/2025-06-18/`

[17] A2A Protocol Project, "Agent2Agent Protocol Specification, version 1.0.0," 2026. `https://a2a-protocol.org/dev/specification/`

[18] GitHub, "A simpler API for authoring commits," GitHub Changelog, Sept. 13, 2021.

[19] Khronos Group, "glTF 2.0 Specification," 2017–2026. `https://github.com/KhronosGroup/glTF/tree/main/specification/2.0`

[20] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3D Gaussian Splatting for Real-Time Radiance Field Rendering," *ACM Transactions on Graphics*, vol. 42, no. 4, art. 139, 2023.

[21] Khronos Group, "KHR_gaussian_splatting glTF Extension," Release Candidate, 2026. `https://github.com/KhronosGroup/glTF/tree/main/extensions/2.0/Khronos/KHR_gaussian_splatting`

[22] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, 2023.

[23] L. Lamport, "Time, clocks, and the ordering of events in a distributed system," *Communications of the ACM*, vol. 21, no. 7, pp. 558–565, 1978.

[24] R. C. Merkle, "A digital signature based on a conventional encryption function," in *CRYPTO '87*, 1987.

[25] J. H. Saltzer and M. D. Schroeder, "The protection of information in computer systems," *Proceedings of the IEEE*, vol. 63, no. 9, pp. 1278–1308, 1975.

[26] W3C, "PROV-O: The PROV Ontology," W3C Recommendation, Apr. 2013.

[27] R. T. Fielding, "Architectural styles and the design of network-based software architectures," Ph.D. dissertation, University of California, Irvine, 2000.

[28] C. A. Ellis and S. J. Gibbs, "Concurrency control in groupware systems," in *ACM SIGMOD*, 1989.